

Continued fractions

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1 Introduction

Continued fractions play a pivotal role in Diophantine approximation. Indeed, a classical result due to Lagrange states that the convergents of a real number are the best possible rational approximations. The first appearance of continued fractions dates back to 1656 when Wallis computed the continued fraction expansion of $4/\pi$ [1]. However, it was Euler who first formalized the theory, most notably in 1744 in *De fractionibus continuis* [2].

Beyond optimal rational approximation, continued fractions encode deeper arithmetic information. In particular, Lagrange’s celebrated theorem [3] characterizes quadratic irrationals by the periodicity of their continued fraction expansion. Thus, continued fractions are not merely an alternative to decimal representation but a bridge between Diophantine approximation and algebraic number theory. Our first goal in this report is to present this connection, in particular through an application to Pell’s equation. To further illustrate the breadth of the subject, we will also describe a geometric — hyperbolic — algorithm for computing continued fractions.

We then turn to Khintchine’s theorem [4], which provides a zero–one law for Diophantine approximation and offers a probabilistic complement to the

deterministic picture given by Lagrange. We will highlight how this theorem is linked to the ergodic properties of the Gauss map, thereby extending the range of mathematical areas naturally connected to continued fractions.

2 Periodic continued fractions

In this section, we introduce Lagrange and Galois's theorems on continued fractions, and some of their consequences, using the book of Rockett and Szűsz [5].

2.1 Quadratic irrational numbers

Definition 2.1.1 (quadratic irrational). A number is a quadratic irrational if it is solution of a quadratic equation $ax^2 + bx + c = 0$, with $a \in \mathbf{Z}^*$, $b, c \in \mathbf{Z}$, and $D = b^2 - 4ac$ is not a perfect square. If $t = t_1 + \sqrt{D}t_2$ (with $t_1, t_2 \in \mathbf{Q}$), we note $\bar{t} = t_1 - \sqrt{D}t_2$.

Notation 2.1.2. Let $x = [a_0; a_1, a_2, \dots]$ be a continued fraction. For all $i \in \mathbf{N}$, we note $x_i = [a_i; a_{i+1}, a_{i+2}, \dots]$. We define $(A_n)_{n \in \mathbf{Z}_{>-2}}$, and $(B_n)_{n \in \mathbf{Z}_{>-2}}$ by $A_{-1} = 1$, $A_0 = 0$, $B_{-1} = 0$, $B_0 = 1$ and the induction relations:

$$\begin{cases} A_{n+1} = a_{n+1}A_n + A_{n-1} \\ B_{n+1} = a_{n+1}B_n + B_{n-1} \end{cases}.$$

Proposition 2.1.3. *For any integer n , we have:*

$$[a_0; a_1, \dots, a_n] = \frac{A_n}{B_n}.$$

Proof. By induction on n . We have that $\frac{A_0}{B_0} = a_0$ and $\frac{A_1}{B_1} = \frac{a_1 a_0 + 1}{a_1} = [a_0; a_1]$. Suppose that the proposition holds for all $k \leq n$. We have

$$[a_0, a_1, \dots, a_n, a_{n+1}] = [a_0, a_1, \dots, a_n + \frac{1}{a_{n+1}}] = \frac{(a_n + \frac{1}{a_{n+1}})A_{n-1} + A_{n-2}}{(a_n + \frac{1}{a_{n+1}})B_{n-1} + B_{n-2}}$$

$$[a_0, a_1, \dots, a_n, a_{n+1}] = \frac{A_n + \frac{1}{a_{n+1}}A_{n-1}}{B_n + \frac{1}{a_{n+1}}B_{n-1}} = \frac{A_{n+1}}{B_{n+1}}.$$

□

Proposition 2.1.4. *For any integer n , we have:*

$$x = \frac{A_n x_{n+1} + A_{n-1}}{B_n x_{n+1} + B_{n-1}}.$$

Proof. By induction on n . We check that

$$\frac{A_0x_1 + A_{-1}}{B_0x_1 + B_{-1}} = \frac{a_0x_1 + 1}{x_1} = a_0 + \frac{1}{x_1} = x.$$

Then, assuming the proposition holds for $n \in \mathbf{N}$, we have

$$\begin{aligned} x &= \frac{A_nx_{n+1} + A_{n-1}}{B_nx_{n+1} + B_{n-1}} = \frac{A_n(a_{n+1} + \frac{1}{x_{n+2}}) + A_{n-1}}{B_n(a_{n+1} + \frac{1}{x_{n+2}}) + B_{n-1}} \\ x &= \frac{a_{n+1}A_n + A_{n-1} + \frac{1}{x_{n+2}}A_n}{a_{n+1}B_n + B_{n-1} + \frac{1}{x_{n+2}}B_n} = \frac{A_{n+1}x_{n+2} + A_n}{B_{n+1}x_{n+2} + B_n}. \end{aligned}$$

□

Definition 2.1.5 (periodic continued fraction). We say that $x = [a_0; a_1, a_2, \dots]$ is periodic if there exists $l \in \mathbf{N}, m \in \mathbf{N}^*$ such that for all $n \in \mathbf{N}$,

$$a_{l+nm} = a_{l+m}, a_{l+nm+1} = a_{l+m+1}, \dots, a_{l+(n+1)m-1} = a_{l+2m-1}.$$

We say that x is purely periodic if x is periodic and $l = 0$. In this case, we note $x = [a_0; a_1, \dots, a_m]$.

Theorem 2.1.6 (Lagrange). $x \in \mathbf{R}$ is a quadratic irrational if and only if x is a periodic continued fraction.

Proof. Let us assume that x is a periodic continued fraction. If x is purely periodic, then $\exists m \in \mathbf{N}, \forall n \in \mathbf{N}, x = x_{nm}$. By Proposition 2.1.4,

$$x = \frac{A_{m-1}x + A_{m-2}}{B_{m-1}x + B_{m-2}}$$

so that

$$B_{m-1}x^2 + (B_{m-2} - A_{m-1})x - A_{m-2} = 0. \quad (1)$$

Since $m > 0$, $B_{m-1} \neq 0$ and x is a quadratic irrational. If x is not purely periodic,

$$\exists l, m \in \mathbf{N}^*, \forall n \in \mathbf{N}, x_l = x_{l+nm}.$$

We can then use Proposition 2.1.4 twice to obtain

$$x = \frac{A_{l-1}x_l + A_{l-2}}{B_{l-1}x_l + B_{l-2}} = \frac{A_{l+m-1}x_l + A_{l+m-2}}{B_{l+m-1}x_l + B_{l+m-2}}.$$

We deduce that

$$x_l = -\frac{B_{l-2}x - A_{l-2}}{B_{l-1}x - A_{l-1}} = -\frac{B_{l+m-2}x - A_{l+m-2}}{B_{l+m-1}x - A_{l+m-1}}.$$

We can then obtain the following equation:

$$\begin{aligned} & (B_{l-2}B_{l+m-1} - B_{l-1}B_{l+m-2})x^2 \\ & + (A_{l-1}B_{l+m-2} + B_{l-1}A_{l+m-2} - B_{l-2}A_{l+m-1} - A_{l-2}B_{l+m-1})x \\ & + A_{l-2}A_{l+m-1} + A_{l-1}A_{l+m-2} = 0. \end{aligned}$$

If x were not a quadratic irrational, it would force $B_{l-2}B_{l+m-1} = B_{l-1}B_{l+m-2}$. This is impossible since B_{l+m-1} does not divide $B_{l-1}B_{l+m-2}$: B_{l+m-1} and B_{l+m-2} are relatively prime, and $B_{l+m-1} > B_{l-1}$. In every case, x is a quadratic irrational.

Now, we assume that x is a quadratic irrational, and we want to prove that x is a periodic continued fraction.

By the quadratic formula, $x = \frac{P_0 + \sqrt{D}}{Q_0}$ with P_0, Q_0 , and D integers, where D is not a perfect square and $Q_0 \neq 0$ and divides $D - P_0^2$. We find by induction that $\forall n \in \mathbf{N}$, $x_n = \frac{P_n + \sqrt{D}}{Q_n}$, by setting $\forall n \in \mathbf{N}^*, P_{n+1} = a_n Q_n - P_n$ and $Q_{n+1} = \frac{D - P_{n+1}^2}{Q_n}$. We then deduce that $\bar{x}_n = \frac{P_n - \sqrt{D}}{Q_n}$. By Proposition 2.1.4, we find that

$$\forall n \in \mathbf{N}, \bar{x} = \frac{A_n \bar{x}_{n+1} + A_{n-1}}{B_n \bar{x}_{n+1} + B_{n-1}}.$$

We deduce that

$$\bar{x}_{n+1} = -\frac{B_{n-1}\bar{x} - A_{n-1}}{B_n\bar{x} - A_n} = -\frac{B_{n-1}}{B_n} \frac{\bar{x} - A_{n-1}/B_{n-1}}{\bar{x} - A_n/B_n}.$$

We deduce by Proposition 2.1.3, that for n large enough, $-1 < \bar{x}_n < 0$. Since $\forall n \in \mathbf{N}^*, x_n > 1$, we deduce that for n large enough, $x_n - \bar{x}_n > 0$, i.e. $\frac{2\sqrt{D}}{Q_n} > 0$, so $Q_n > 0$. We also have $\bar{x}_n + x_n > 0$, i.e. $\frac{2P_n}{Q_n} > 0$, so $P_n > 0$. Because $\bar{x}_n < 0$, we deduce that $P_n < \sqrt{D}$. But because $\bar{x}_n > -1$, $\sqrt{D} - P_n < Q_n$. We also deduce from $x_n > 1$ that $Q_n < \sqrt{D} + P_n$.

We conclude that for n large enough,

$$0 < P_n < \sqrt{D} \text{ and } 0 < \sqrt{D} - P_n < Q_n < \sqrt{D} + P_n.$$

There are only finitely many pairs (P_n, Q_n) , so the set $\{x_n, n \in \mathbf{N}\}$ is finite. But by construction of a continued fraction, if we set $F: x \mapsto \frac{1}{x - [x]}$, we have $F(x_n) = x_{n+1}$. We deduce that the sequence $(x_n)_{n \in \mathbf{N}}$ is ultimately periodic, so that x is a periodic continued fraction. $\square$

Example 2.1.7. This gives us a way to compute continued fraction expansions for quadratic irrationals using the sequence $(P_n, Q_n)_{n \in \mathbf{N}}$. For example, if we take $x = \frac{7 + \sqrt{11}}{19}$, we can take $Q_0 = 19$, $P_0 = 7$ and $D = 11$, since $19 \mid 11 - 7^2$. $[x] = a_0 = 0$, and so $P_1 = 0 \times 19 - 7 = -7$, $Q_1 = \frac{11 - (-7)^2}{19} = -2$, and $x_1 = \frac{-7 + \sqrt{11}}{-2}$, so $a_1 = 1$. Repeating this process, we find $P_2 = 5$, $Q_2 = 7$ and $a_2 = 1$, $P_3 = 2$, $Q_3 = 1$, $a_3 = 5$, $P_4 = 3$, $Q_4 = 2$, and $a_4 = 3$. Because

$x_4 > 1$ and $-1 < \overline{x_4} < 0$, the periodic block starts there. We have $P_5 = 3$, $Q_5 = 1$, $a_6 = 6$ and $P_6 = 3$, $Q_6 = 2$ and $a_6 = 3$. So $x_6 = x_4$ and we are done ; $\frac{7+\sqrt{11}}{19} = [0; 1, 1, 5, \overline{3, 6}]$.

Definition 2.1.8 (reduced). Let x be a quadratic irrational. x is called reduced if $x > 1$ and $0 > \overline{x} > -1$.

Theorem 2.1.9 (Galois). Let x be a quadratic irrational. x is purely periodic if and only if x is reduced.

Proof. Suppose that $x = [a_0; a_1, \dots]$ is purely periodic. Since the value of a_0 appears again as a_m , for some $m \geq 1$, we deduce that $a_0 \geq 1$, so $x \geq 1$. Let

$$f(t) = B_{m-1}t^2 + (B_{m-2} - A_{m-1})x - A_{m-2}.$$

By eq. (1) in the proof of Lagrange's theorem, $f(x) = 0$. Since

$$f(0) = -A_{m-2} < 0 \text{ and } f(-1) = (B_{m-1} - B_{m-2}) + (A_{m-1} - A_{m-2}) > 0,$$

the other root of f must be between -1 and 0 . But this other root is $\overline{x}$, so x is reduced.

Now let us assume that x is reduced. We find by induction that $-1 < \overline{x_n} < 0$ for all $n \in \mathbb{N}$. By contradiction, we assume that x is not purely periodic. Then $a_{l-1} \neq a_{l+m-1}$. However, $x_l = x_{l+m}$ and

$$x_{l-1} - x_{l+m-1} = (a_{l-1} + \frac{1}{a_l}) - (a_{l+m-1} + \frac{1}{x_{l+m}}) = a_{l-1} - a_{l+m-1}$$

is a non-zero integer. This is impossible because $-1 < x_{l-1} - x_{l+m-1} < 1$. $\square$

Example 2.1.10. For example, if we take $x = \frac{1+\sqrt{3}}{2}$, then x is reduced, so by the preceding theorem it is purely periodic, and indeed, $x = [1, 2]$.

Proposition 2.1.11. Let $x = [\overline{a_0}; \overline{a_1}, \dots, \overline{a_{m-1}}]$ be a non-zero real number. Then the continued fraction of $-\frac{1}{\overline{x}}$ is given by $-\frac{1}{\overline{x}} = [\overline{a_{m-1}}; \overline{a_{m-2}}, \dots, \overline{a_0}]$.

Proof. We have

$$x = a_0 + \frac{1}{x_1}, x_1 = a_1 + \frac{1}{x_2}, \dots, x_{m-1} = a_{m-1} + \frac{1}{x}.$$

Conjugation gives

$$\overline{x} = a_0 + \frac{1}{\overline{x_1}}, \dots, \overline{x_{m-1}} = a_{m-1} + \frac{1}{\overline{x}}.$$

We deduce that

$$-\frac{1}{\overline{x}} = a_{m-1} - \overline{x_{m-1}}, \dots, -\frac{1}{\overline{x_1}} = a_0 - \overline{x},$$

which gives us the result intended. $\square$

Proposition 2.1.12. *Let $d > 1$ be a rational number that is not the square of another rational number. Then $\sqrt{d} = [a_0; a_1, a_2, \dots, a_2, a_1, 2a_0]$.*

Proof. Let $x = \sqrt{d}$. We have $x_1 = \frac{1}{x - [x]}$, so $\overline{x_1} = -\frac{1}{x + [x]}$, and x_1 is reduced. By Galois's theorem, x_1 is purely periodic, let's say $x_1 = [\overline{a_1; a_2, \dots, a_{m-1}}]$. By the preceding proposition, $-\frac{1}{x_1} = [\overline{a_{m-1}; a_{m-2}, \dots, a_1}]$. We then have

$$[2a_0, \overline{a_1, \dots, a_{m-1}}] = a_0 + x = [x] + x = -\frac{1}{x_1} = [a_{m-1}, \overline{a_{m-2}, \dots, a_1, a_{m-1}}]$$

and the result is proved. $\square$

Proposition 2.1.13. *Let $d > 1$ be an integer that is not a perfect square, and let m be the period of the continued fraction for $\sqrt{d}$. Then, with the preceding notation, $\forall n \in \mathbf{N}$, $Q_n = 1$ if and only if $m|n$.*

Proof. We have $x_1 = x_{m+1}$, while $x_0 = \sqrt{d}, \dots, x_m$ are all distinct. Thus $x_m = 2a_0 + \frac{1}{x_{m+1}} = 2a_0 + \frac{1}{x_1} = a_0 + \sqrt{d}$. But $x_m = \frac{P_m + \sqrt{d}}{Q_m}$. If Q_m were strictly greater than 1, we would have $P_m - a_0 Q_m = (Q_m - 1)\sqrt{d}$, and $P_m - a_0 Q_m$ is rational, while $(Q_m - 1)\sqrt{d}$ would not. Thus $Q_m = 1$. A similar argument shows that $Q_n = 1$ for every n multiple of m .

Now assume that $Q_n = 1$, for some $n \in \mathbf{N}^*$. Then $x_n = P_n + \sqrt{d}$. Since $n \geq 1$, x_n is purely periodic and we deduce that $0 < \sqrt{d} - P_n < 1$, i.e., $P_n < \sqrt{d} < P_n + 1$, so that $P_n = [\sqrt{d}] = a_0$ and $x_n = x_m$ and n is a multiple of m . $\square$

2.2 Pell's equation

In this section, we give an application of the study of periodic continued fractions to Pell's equation:

$$x^2 - dy^2 = N \tag{2}$$

where N is an integer and d is not a perfect square. Moreover, we denote by $(A_k/B_k)_{k \in \mathbf{N}}$ the sequence of convergents of $\sqrt{d}$.

Theorem 2.2.1. *If $|N| < \sqrt{d}$ then the relatively prime solutions of eq. (2) are given by $(x, y) = (A_k, B_k)$ for an integer k .*

Proof. We first recall a classical result :

Theorem 2.2.2 (Legendre). *If $x/y \in \mathbf{Q}$ is an irreducible fraction and $\alpha \in \mathbf{R} \setminus \mathbf{Q}$ such that $\left| \frac{x}{y} - \alpha \right| < \frac{1}{2y^2}$, then x/y is a convergent of α .*

Let (x, y) be a relatively prime solution of eq. (2). If $N > 0$, we can assume that $x, y > 0$ because if (x, y) is a solution, so is $(\pm x, \pm y)$.

We write $(x + \sqrt{d}y)(x - \sqrt{d}y) = N > 0$, so because $x + \sqrt{d}y > 0$, $x - \sqrt{d}y > 0$ so $x > \sqrt{d}y$.

$$\left| \sqrt{d} - \frac{x}{y} \right| = \left| \frac{dy^2 - x^2}{y(x + \sqrt{d}y)} \right| = \frac{N}{y(x + \sqrt{d}y)} < \frac{N}{2\sqrt{d}y^2} < \frac{1}{2y^2}.$$

Because (x, y) is relatively prime, x/y is irreducible, and we can apply Legendre's theorem to obtain the desired result.

If $N < 0$, we rewrite eq. (2) as $y^2 - \frac{1}{d}x^2 = \frac{-N}{d} > 0$ and we use a similar line of reasoning to find that y/x is a convergent of $1/\sqrt{d}$, which gives us the desired result. $\square$

Remark. This gives us a way to compute units of quadratic fields. Indeed, let $K = \mathbf{Q}(\sqrt{d})$, $\mathcal{O}_K$ be its ring of integers, with d square-free. If $d \equiv 2, 3[4]$, then $\mathcal{O}_K = \mathbf{Z}[\sqrt{d}]$, and a unit is of the form $u = x + \sqrt{d}y$, with $u^{-1} \in \mathbf{Z}[\sqrt{d}]$. But we have $u^{-1} = \frac{x - \sqrt{d}y}{x^2 - dy^2}$. This implies $x^2 - dy^2 = \pm 1$, and we deduce that a unit is of the form $A_k + \sqrt{d}B_k$, with (A_k, B_k) verifying the Pell equation with $N = \pm 1$.

If $d \equiv 1[4]$, then $\mathcal{O}_K = \mathbf{Z}[\frac{1+\sqrt{d}}{2}]$, and by a similar method we find that a unit is of the form $A_k + \frac{1+\sqrt{d}}{2}B_k$, with (A_k, B_k) verifying the Pell equation with $N = \pm 4$.

Theorem 2.2.3. *If $N = 1$, then eq. (2) is always solvable and if $N = -1$, then eq. (2) is solvable if and only if the period of the continued fraction of $\sqrt{d}$ is odd.*

Proof. Let us write $\sqrt{d} = [a_0; a_1, \dots]$. For any natural integer i , let us write $x_i = [a_i; a_{i+1}, \dots] = \frac{P_i + \sqrt{d}}{Q_i}$. Let us fix $i \in \mathbf{N}$. By Proposition 2.1.4 we have

$$\sqrt{d} = \frac{A_i x_{i+1} + A_{i-1}}{B_i x_{i+1} + B_{i-1}} = \frac{A_i(P_{i+1} + \sqrt{d}) + A_{i-1}Q_{i+1}}{B_i(P_{i+1} + \sqrt{d}) + B_{i-1}Q_{i+1}}.$$

We deduce that $A_i = B_i P_{i+1} + B_{i-1} Q_{i+1}$ and $d B_i = A_i P_{i+1} + A_{i-1} Q_{i+1}$. We then compute $A_i^2 - d B_i^2$ to find that it is equal to $(-1)^{i+1} Q_{i+1}$. By the preceding theorem, the only possible solutions to Pell's equation are the $(-1)^i Q_i$, with $i \in \mathbf{N}^*$. Furthermore, by Proposition 2.1.13, $Q_i = 1$ if and only if $m|i$. That gives us the desired result. $\square$

Example 2.2.4. For instance, $x^2 - 29y^2 = 1$ and $x^2 - 29y^2 = -1$ are both solvable because the period of the continued fraction associated to $\sqrt{29}$ is odd. On the other hand, $x^2 - 34y^2 = 1$ is solvable while $x^2 - 34y^2 = -1$ is not.

2.3 Geometric interpretation

In this section, we outline how one can geometrically recover the continued fraction of a real, following the work of Series [6].

Let us denote by $\mathbf{H}$ the Poincaré upper half plane $\{z \in \mathbf{C}, \Im(z) > 0\}$. Then, $\text{SL}_2(\mathbf{R})$ acts on $\mathbf{H}$ by

$$\forall z \in \mathbf{H}, \begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az + b}{cz + d}$$

where the above matrix belongs to $\text{SL}_2(\mathbf{R})$.

The Poincaré plane $\mathbf{H}$ is endowed with the hyperbolic metric $\frac{dx^2 + dy^2}{y^2}$. For this metric, the geodesics of $\mathbf{H}$ are exactly the semi-circles and the vertical

lines. Moreover, the isometries of $\mathbf{H}$ are exactly the one given by the action of $\mathrm{SL}_2(\mathbf{R})$ on $\mathbf{H}$. In particular, $\mathrm{SL}_2(\mathbf{Z})$ acts by isometries on $\mathbf{H}$. This can be applied to continued fractions. Indeed, let us denote $z \mapsto z + a$ by T^a for any real a and $z \mapsto 1/z$ by S . Then, for any real $\xi = [a_0, a_1 \dots]$, we can notice that the n th convergent of ξ is given by $T^{a_0} S T^{a_1} \dots S T^{a_{n-1}} S(a_n)$. Hence, one might naturally want to build a correspondence between continued fractions and geodesics. In order to do so, we need to tessellate $\mathbf{H}$.

Definition 2.3.1 (Farey tessellation). The Farey tessellation of $\mathbf{H}$ is the set:

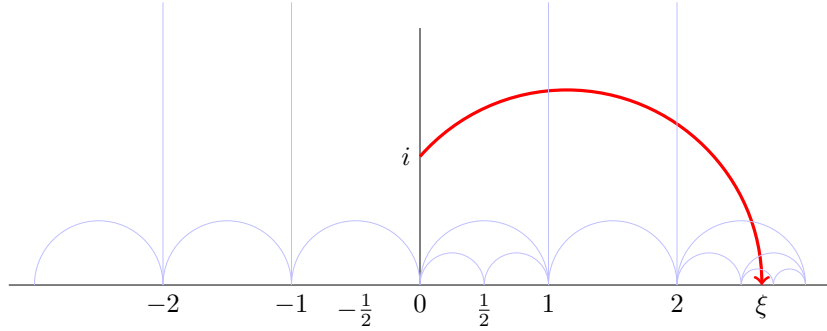
$$\mathcal{F} = \bigcup_{A \in \mathrm{SL}_2(\mathbf{Z})} A \cdot i\mathbf{R}_{>0},$$

where $i\mathbf{R}_{>0}$ is oriented from ∞ to 0.

Then, applying T^k to $i\mathbf{R}_{>0}$ on $i\mathbf{R}_{>0}$ for any integer k , we can notice that $\mathcal{F}$ contains all vertical lines starting from an integer. Moreover, applying the usual rules of calculus on $\mathbf{H} \cup \mathbf{R} \cup \{\infty\}$, we have:

$$\forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbf{Z}), \frac{a \times \infty + b}{c \times \infty + d} = \frac{a}{c} \text{ and } \frac{a \times 0 + b}{c \times 0 + d} = \frac{b}{d}.$$

Hence, the geodesics of $\mathcal{F}$ that are not vertical lines are exactly the semi-circles from a rational $\frac{a}{b}$ to a rational $\frac{c}{d}$ such that $ad - bc = 1$. On the figure below, we have represented in light blue some geodesics from $\mathcal{F}$.



Now, let us fix a real $\xi = [a_0, a_1 \dots]$. Let us consider the oriented geodesic γ going from i to ξ as on the figure above (in red). Then, γ intersects $\mathcal{F}$ several times. First, consider the hyperbolic triangle whose vertices are 0, 1 and ∞ . Since the geodesic γ is oriented, it is convenient to talk about the sides $[0, 1]$ as the side to the right, and of $[1, \infty]$ as the side to the left of $[0, \infty]$. In other words, we can say that γ goes out from the first tessell from the side to the left of the side it began — $[0, \infty]$. Then, it enters the second tessell which is a hyperbolic triangle of vertices 1, 2 and ∞ . Here, γ enters from the side $[1, \infty]$ and goes out from the side to the left i.e., $[2, \infty]$. On the opposite, γ enters the hyperbolic triangle $[2, 3, \infty]$ from $[2, \infty]$ and goes out from the side to the right

i.e., [2, 3]. Now, we can associate to γ the sequence $(b_i)_{i \in \mathbf{N}}$ of integers such that γ goes out from the left side b_0 times, then goes out from the right side b_1 times, then goes out from the left side b_2 times and so on. So, on the figure above, the first terms of the sequence associated to γ would be 2, 1, 1, ...

Theorem 2.3.2. *With the above construction, $b_i = a_i$ for any integer i .*

In other words, one can recover the continued fraction of ξ using the above algorithm.

The idea behind this construction is that the first time γ goes out from the side to the right corresponds to the entire part of ξ . This can be seen from the figure above. Then, we apply ST^{-a_0} to the figure, and we get the same tessellation $\mathcal{F}$ and ξ has been replaced by $\frac{1}{x-a_0}$. Hence, if we apply the process again, we should obtain a_1 . Moreover, since we applied S , the left and right sides exchanged and therefore, we count the number of times γ goes out from the side to the right when computing a_1 .

This construction can be generalized to a geodesic between two real numbers. In particular, we can give a geometric analogue of Theorem 2.1.6 due to Sarnak [7].

Theorem 2.3.3 (Sarnak). *Let α and β be two real numbers and let γ be a geodesic from α to β . Then, α and β are conjugate real quadratic irrational numbers if and only if the projection of γ in $\mathbf{H} \backslash \mathrm{PSL}_2(\mathbf{Z})$ is a closed geodesic.*

Hence, the periodicity of continued fractions is geometrically translated by the periodicity of geodesics.

3 Khintchine's theorem

In this section, we introduce Khintchine's theorem and its consequences. Following the work of Bugeaud [8], we use the continued fractions theory to give a sketch proof.

3.1 The theorem

Throughout this section, λ denotes the Lebesgue measure.

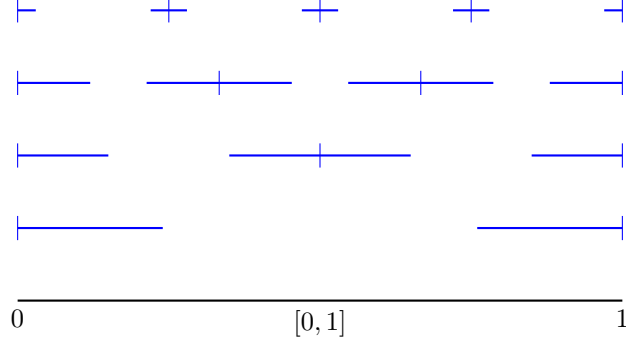
Theorem 3.1.1 (Khintchine). *Let $\Psi: \mathbf{R}_{\geq 1} \rightarrow \mathbf{R}_{>0}$ be a continuous function. Then, the set*

$$\mathcal{K}_1^*(\Psi) = \left\{ \xi \in \mathbf{R}, \left| \xi - \frac{p}{q} \right| < \Psi(q) \text{ for infinitely many reduced } \frac{p}{q} \in \mathbf{Q} \right\}$$

has Lebesgue measure zero if $\sum_{q \in \mathbf{N}^} q\Psi(q)$ converges and has full Lebesgue measure if $x \mapsto x^2\Psi(x)$ is non-increasing and $\sum_{q \in \mathbf{N}^*} q\Psi(q)$ diverges.*

Let us guess why this theorem should hold. Let us focus on $[0, 1]$. Let Ψ be a map as in Theorem 3.1.1. First, we draw around 0 and 1 segments of

length $2\Psi(1)$ and we intersect them with $[0, 1]$. Then, we draw around $0, \frac{1}{2}$ and 1 segments of length $2\Psi(2)$ intersected with $[0, 1]$. At step n , we draw around $0, \frac{1}{n}, \frac{2}{n}, \dots, 1$ segments of length $2\Psi(n)$ intersected with $[0, 1]$. Here is an illustration of step 4:



Now, a real number ξ belongs to $\mathcal{K}_1^*(\Psi)$ if and only if it belongs to infinitely many of the above intervals. Therefore, since the Lebesgue measure of intervals drawn at step n is less or equal to $n\Psi(n)$, the convergence half of Theorem 3.1.1 is a direct consequence of the so-called Borel-Cantelli Lemma:

Lemma 3.1.2 (Borel-Cantelli). *Let $(E_n)_{n \in \mathbf{N}}$ be a sequence of Borelian real subsets such that $\sum_{n \in \mathbf{N}} \lambda(E_n)$ converges. Then*

$$\lambda \left(\limsup_{n \in \mathbf{N}} E_n \right) = 0.$$

Remark. In Lemma 3.1.2 above, $\limsup_{n \in \mathbf{N}} E_n$ is the subset of real numbers belonging to an infinity of sets E_n .

The divergence half of Theorem 3.1.1 is much harder and cannot be deduced from Lemma 3.1.2. To prove the divergence half, we need two theorems.

Theorem 3.1.3 (Borel-Bernstein). *Let $(u_n)_{n \in \mathbf{N}^*}$ be a sequence of positive real numbers. If $\sum_{n \in \mathbf{N}^*} u_n^{-1}$ diverges, then, for almost all $[0; a_1, a_2, \dots] \in [0, 1[$, there exist infinitely many integers n such that $a_n \geq u_n$. Otherwise, for almost all $[0; a_1, a_2, \dots] \in [0, 1[$, there exist only a finite number of integers n such that $a_n \geq u_n$.*

Theorem 3.1.4 (Lévy). *There exists a positive real number β such that for almost all real ξ , there exists an integer $n_0(\xi)$ such that $q_n < e^{\beta n}$ for all integers $n \geq n_0(\xi)$, where $\frac{p_n}{q_n}$ denotes the n th convergent of ξ .*

Theorem 3.1.4 obviously follows from the following theorem:

Theorem 3.1.5 (Lévy's constant). *For almost any real number ξ whose sequence of reduced convergents is denoted by $(p_n/q_n)_{n \in \mathbf{N}}$, we have:*

$$\lim_{n \rightarrow \infty} \sqrt[n]{q_n} = \exp \left(\frac{\pi^2}{12 \ln(2)} \right).$$

Theorem 3.1.5 is a consequence of ergodic theory. In particular, it can be proved thanks to Birkhoff's ergodic theorem [9].

Definition 3.1.6 (ergodic transformation). Let (X, Σ, μ) be a probability space and let $T: X \rightarrow X$ be a measure-preserving transformation. We say that T is ergodic if for any $E \in \Sigma$ that is invariant i.e., such that $\mu(T^{-1}(E) \Delta E) = 0$, either $\mu(E) = 0$ or $\mu(E) = 1$.

Definition 3.1.7 (time average). Let (X, Σ, μ) be a measure space, $T: X \rightarrow X$ a measurable transformation and let $f \in L^1(\mu)$. We define the time average of f (associated to T) as the function $\hat{f}$ (possibly partially defined) such that:

$$\forall x \in X, \hat{f}(x) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} f(T^k(x)).$$

Definition 3.1.8 (space average). Let (X, Σ, μ) be a measure space such that μ is finite and non-zero, $T: X \rightarrow X$ be a measurable transformation and let $f \in L^1(\mu)$. We define the space average of f (associated to T) as the function $\bar{f}$ such that:

$$\forall x \in X, \bar{f}(x) = \int f d\mu.$$

Theorem 3.1.9 (Birkhoff's ergodic theorem). *Let (X, Σ, μ) be a probability space. Let $T: X \rightarrow X$ be an ergodic transformation. Then, the time and space averages of any $f \in L^1(\mu)$ associated to T coincide.*

Now, notice that in Definition 3.1.7, the operator T is recursively applied to a point of X . Thanks to the recursive definition of continued fractions, this observation can be used to prove Theorem 3.1.5.

Definition 3.1.10 (Gauss map). The Gauss map is defined as:

$$T: \begin{array}{ccc} (0, 1) \setminus \mathbf{Q} & \longrightarrow & (0, 1) \setminus \mathbf{Q} \\ x & \longmapsto & \frac{1}{x} - \lfloor \frac{1}{x} \rfloor \end{array}$$

Then, it is known that the Gauss map is ergodic for the Gauss distribution $\frac{dt}{(1+t) \ln(2)}$ (see for instance Cornfeld, Fomin and Sinai [10]).

Before proving Theorem 3.1.5, we need a lemma.

Lemma 3.1.11. *For any $\xi \in (0, 1)$, we have:*

$$\forall n \in \mathbf{N}, \left| \ln(\xi) - \ln\left(\frac{p_n}{q_n}\right) \right| < \frac{1}{\varphi_n}$$

where $(p_n/q_n)_{n \in \mathbf{N}}$ denotes the sequence of convergents of ξ and $(\varphi_n)_{n \in \mathbf{N}}$ denotes the Fibonacci sequence.

Proof. Let $\xi \in (0, 1)$, $n \in \mathbf{N}$ and let us denote by p_{n-1}/q_{n-1} and p_n/q_n its $(n-1)$ th and n th convergents. Let us consider the function $f: t \mapsto \ln \frac{p_n + p_{n-1}t}{q_n + q_{n-1}t}$.

Notice that $f(0) = \ln(p_n/q_n)$ and $f(T^n(x)) = \ln(x)$. By the mean value theorem, we have for any $t \in [0, 1]$:

$$|f(t) - f(0)| = \max_{t \in [0, 1]} \frac{1}{(p_n + p_{n-1}t)(q_n + q_{n-1}t)} = \frac{1}{p_n q_n} \leq \frac{1}{q_n} \leq \frac{1}{\varphi_n}$$

where the last inequality follows from the recursive relation defining the sequence of denominators. $\square$

Using this lemma, we can prove Theorem 3.1.5. Without loss of generality, we can show the result on $(0, 1)$.

Proof of Theorem 3.1.5. Let $\xi \in (0, 1)$. We denote by $(p_n(x)/q_n(x))_{n \in \mathbf{N}}$ the convergents sequence of any real x .

Notice that $p_n(\xi)$ is equal to $q_{n-1}(T\xi)$ for any positive integer n . Therefore, we have:

$$-\ln(q_n(\xi)) = \ln \frac{p_n(\xi)}{q_n(\xi)} + \ln \frac{p_{n-1}(T\xi)}{q_{n-1}(T\xi)} + \cdots + \ln \frac{p_1(T^{n-1}\xi)}{q_1(T^{n-1}\xi)}.$$

By Lemma 3.1.11, there is a real δ such that

$$\delta \leq \sum_{k=1}^{\infty} \frac{1}{\varphi_k} < \infty$$

and

$$-\ln(q_n(\xi)) = \ln(\xi) + \ln(T\xi) + \cdots + \ln(T^{n-1}\xi) + \delta.$$

Hence, we deduce from Theorem 3.1.9 that for almost any $x \in (0, 1)$

$$\lim_{q \rightarrow \infty} \frac{\ln q_n(x)}{n} = \int_0^1 -\frac{\ln(t)}{(1+t \ln(2))} dt = \frac{\pi^2}{12 \ln(2)}.$$

This concludes the proof. $\square$

Now, we can finally prove Theorem 3.1.1.

Proof of the divergence half of Theorem 3.1.1. Let us consider a map Ψ as in Theorem 3.1.1. Moreover, we suppose that $\sum_{q \in \mathbf{N}^*} q\Psi(q)$ diverges. As in the convergence half proof, it is sufficient to prove that $\mathcal{K}_1^*(\Psi) \cap [0, 1]$ has full measure. First, let us write

$$\Phi: \begin{array}{ccc} \mathbf{R}_{\geq 1} & \longrightarrow & \mathbf{R}_{>0} \\ x & \longmapsto & e^{2\beta x} \Psi(e^{\beta x}) \end{array}.$$

By hypothesis, Φ is non-increasing. In addition, we have:

$$\forall (a, A) \in \mathbf{R}^2, 1 \leq a < A \implies \int_a^A \Phi(s) ds = \frac{1}{\beta} \int_{e^{\beta a}}^{e^{\beta A}} u \Psi(u) du.$$

By monotony and divergence hypothesis, the right member diverges when A goes to infinity, and hence, so do the left member. Therefore, the monotony implies that $\sum_{q \in \mathbf{N}^*} \Phi(q)$ diverges. Consequently, we can apply Theorem 3.1.3: for almost all real numbers $\xi = [0, a_1, \dots] \in [0, 1]$, whose n th convergent is denoted by q_n , there are infinitely many integers n such that $a_{n+1} \geq \frac{1}{\Phi(n)}$ and so:

$$\left| \xi - \frac{p_n}{q_n} \right| \leq \frac{1}{q_n q_{n+1}} < \frac{1}{a_{n+1} q_n^2} \leq \frac{\Phi(n)}{q_n^2}.$$

Moreover, Theorem 3.1.4 can be reformulated as follows: for almost all real numbers ξ , $n > \frac{\ln(q_n)}{\beta}$ for any integer n large enough, where q_n denotes the n th convergent of ξ . Therefore, since Φ is non-increasing, for almost all $\xi \in [0, 1]$, whose n th convergent is denoted by q_n , there exist infinitely many integers n such that

$$\left| \xi - \frac{p_n}{q_n} \right| < \frac{1}{q_n^2} \Phi \left(\frac{\ln(q_n)}{\beta} \right) = \Psi(q_n).$$

□

3.2 Perspectives

The aim of this section is to give an overview on generalizations of Theorem 3.1.1.

First of all, the conditions of Theorem 3.1.1 might be too strong. Recently, Koukoulopoulos and Maynard [11] proved the Duffin-Schaeffer conjecture which states an analogous of Theorem 3.1.1 with weaker hypothesis.

We denote by φ the Euler totient function.

Theorem 3.2.1 (Koukoulopoulos-Maynard). *Let $\Psi: \mathbf{N} \rightarrow \mathbf{R}_+$ be a function. Then $\mathcal{K}_1^*(\Psi)$ has full Lebesgue measure if $\sum_{q=1}^{+\infty} \varphi(q)\Psi(q)$ diverges.*

Theorem 3.1.1 is one-dimensional since it deals only with real numbers. We can generalize to dimension d by replacing $\sum_{q \in \mathbf{N}^*} q\Psi(q)$ by $\sum_{q \in \mathbf{N}^*} (q\Psi(q))^d$ in Theorem 3.1.1, where Ψ is only asked to be decreasing (see Beresnevich and Velani [12]). Then, we can go even further and ask whether Theorem 3.1.1 can be generalized to manifolds. For instance, Beresnevich and Yang [13] showed the following theorem:

Theorem 3.2.2 (Beresnevich-Yang). *Let $d \geq 2$ be an integer. Let $\mathcal{M}$ be a non-degenerate submanifold (see below) of $\mathbf{R}^d$. Let $\Psi: \mathbf{R}_{>0} \rightarrow (0, 1)$ be a monotonic function. If $\sum_{q \in \mathbf{N}^*} (q\Psi(q))^d$ converges then*

$$\left\{ (\xi_i)_{i=1}^d \in \mathcal{M}, \left| \xi_i - \frac{p_i}{q} \right| < \Psi(q) \text{ for infinitely many } (p_1, \dots, p_n, q) \in \mathbf{Z}^d \times \mathbf{N} \right\}$$

has Lebesgue measure equal to zero.

Definition 3.2.3. Let $d \geq 2$ be an integer. Let $\mathcal{M}$ be a submanifold of $\mathbf{R}^d$. We say that $\mathcal{M}$ is non-degenerate if for any $x \in \mathcal{M}$, there exists a non-degenerate

local chart $f: U \rightarrow \mathcal{M}$ i.e., for almost all $y \in U$, there exists a positive integer l such that f is l times continuously differentiable on a neighborhood of y , and the partial derivatives of f at y of orders up to l span $\mathbf{R}^d$.

3.3 Applications

In this section, we give applications of Theorem 3.1.1 and of its generalizations from the previous section. We aim at showing how useful can these theorems be when computing the Lebesgue measure of some sets.

Let us first give an application of Theorem 3.1.1 to very well approximable numbers.

Definition 3.3.1. A real number ξ is said to be very well approximable if there exists a positive real number τ such that $|\xi - \frac{p}{q}| < \frac{1}{q^{2+\tau}}$ holds for infinitely many rational numbers $\frac{p}{q}$ — in this case, we say that ξ is τ -very well approximable.

Corollary 3.3.2. *The Lebesgue measure of the set of very well approximable numbers is zero.*

Proof. Since a countable union of sets of measure zero is of measure zero, it suffices to notice that

$$\Psi: \begin{array}{ccc} \mathbf{R}_{\geq 1} & \longrightarrow & \mathbf{R}_{>0} \\ x & \longmapsto & \frac{1}{x^{2+k}} \end{array}$$

satisfies the hypothesis from Theorem 3.1.1 for any positive integer k . □

Even though the Lebesgue measure of very well approximable numbers is zero, these are not negligible in a geometric sense. Indeed, their Hausdorff dimension is not trivial (see Beresnevich and Velani [12]). This is the content of the next theorem.

Theorem 3.3.3 (Jarník-Besicovitch). *Let τ be a positive real. The set of real numbers τ -very well approximable has Hausdorff dimension equal to $\frac{2}{\tau+1}$.*

Notice that the set of Liouville numbers i.e., numbers τ -very well approximable for all positive real τ , has both Lebesgue measure and Hausdorff dimension equal to zero.

Now, let us turn to well approximable numbers.

Definition 3.3.4. A real number ξ is said to be badly approximable if there exists a positive real c such that $|\xi - \frac{p}{q}| \geq \frac{c}{q^2}$ for any rational number $\frac{p}{q}$ distinct from ξ . A real number ξ that is not badly approximable is said to be well approximable.

Corollary 3.3.5. *The set of well approximable numbers has full measure.*

Proof. For any positive real c , the function

$$\Psi_c: \begin{array}{ccc} \mathbf{R}_{\geq 1} & \longrightarrow & \mathbf{R}_{>0} \\ x & \longmapsto & \frac{c}{x^2} \end{array}$$

meets the hypothesis from Theorem 3.1.1. Moreover, for all positive real c , $\sum_{q \in \mathbf{N}^*} q \Psi(q) = \sum_{q \in \mathbf{N}^*} \frac{c}{q}$ diverges. Consequently, by Theorem 3.1.1, for all positive real c and for almost all real ξ , there exist infinitely many rational $\frac{p}{q}$ such that $|\xi - \frac{p}{q}| < \frac{2^n}{q^2}$. $\square$

Here again, the Hausdorff dimension gives a contrasting result since the Hausdorff dimension of badly approximable numbers is equal to 1.

Now, let us turn to some applications of Theorem 3.2.1.

Corollary 3.3.6. *For almost any $\alpha \in \mathbf{R}$, there exist infinitely many prime number p and integer m such that $|\alpha - \frac{m}{p}| < \frac{1}{p^2}$.*

Proof. Let us consider $\Psi: \mathbf{N} \rightarrow \mathbf{R}_+$ such that for any integer n , $\Psi(n)$ is equal to $\frac{1}{n^2}$ whenever n is prime and is equal to zero otherwise. Then, notice that for any prime number p , we have:

$$\frac{\varphi(p)}{p^2} = \frac{p-1}{p^2} \geq \frac{1}{2p}.$$

Then, by divergence of $\sum_p \frac{1}{p}$ — where the sum is over prime numbers — and by Theorem 3.2.1, $\mathcal{K}_1^*(\Psi)$ has full Lebesgue measure. $\square$

Corollary 3.3.7. *For almost any $\alpha \in \mathbf{R}$, there exist infinitely many perfect square p^2 and integer m such that $|\alpha - \frac{m}{p^2}| < \frac{1}{p^2}$.*

Proof. Let us consider $\Psi: \mathbf{N} \rightarrow \mathbf{R}_+$ such that for any integer n , $\Psi(n)$ is equal to $\frac{1}{n}$ whenever n is a perfect square and is equal to zero otherwise. Then, notice that for any prime number p , we have:

$$\frac{\varphi(p^2)}{p^2} = \frac{p-1}{p} \geq \frac{1}{2}.$$

Then, by divergence of $\sum_p \frac{1}{2}$ and by Theorem 3.2.1, $\mathcal{K}_1^*(\Psi)$ has full Lebesgue measure. $\square$

Finally, let us see some applications of Theorem 3.2.2.

Corollary 3.3.8. *Let $d \geq 2$ be an integer. Let $\mathcal{M}$ be a non-degenerate submanifold of $\mathbf{R}^d$. Then, for any real $\kappa > \frac{d+1}{d}$ the set*

$$\left\{ (\xi_i)_{i=1}^d \in \mathcal{M}, \left| \xi_i - \frac{p_i}{q} \right| < \frac{1}{q^\kappa} \text{ for infinitely many } (p_1, \dots, p_n, q) \in \mathbf{Z}^d \times \mathbf{N} \right\}$$

has Lebesgue measure equal to zero.

Proof. The proof is similar to Corollary 3.3.2 but the function Ψ from the proof of Corollary 3.3.2 should be replaced by $x \mapsto \frac{1}{x^\kappa}$. $\square$

To give a better understanding of when to apply Corollary 3.3.8, let us explain what are non-degenerate manifolds. Intuitively, a manifold is non-degenerate when it is not locally contained in a hyperplane — so hyperplanes are degenerate for instance. Spheres, torus, paraboloid and Veronese surfaces are all examples of non-degenerate manifolds.

4 Conclusion

Continued fractions stand at the intersection of arithmetic, geometry and dynamics. We saw how periodic expansions characterize quadratic irrationals, allow explicit resolution of Pell-type equations, and correspond to closed geodesics on the modular surface. Khintchine’s theorem then revealed a metric point of view, showing that Diophantine approximation is shaped by metric laws with far-reaching extensions. What begins as a simple iterative process opens onto a surprisingly wide landscape.

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